

# EE-482: Network Security and Cryptography

## LECTURES NO 5:

- i. Symmetric Key Cryptographic Algorithms*
- ii. Algorithm modes*

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# Organization of Lecture

1. Algorithm Types
2. Algorithm Modes

# 1. Algorithm Types (1/6)

Single Key Algorithms:

DES (Data Encryption Standard)

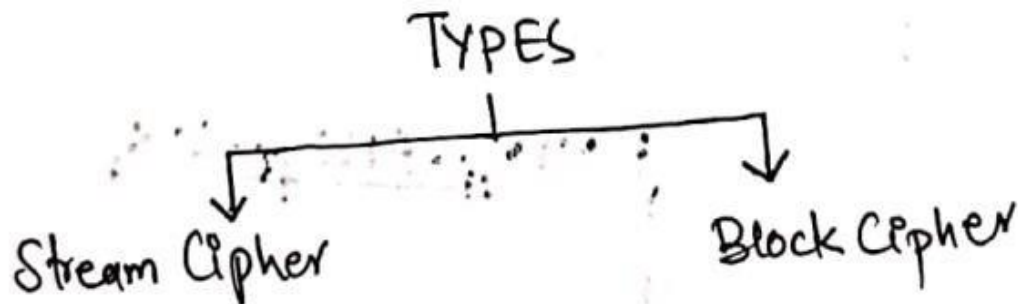
IDEA (International Data Encryption Algorithm)

AES (Advanced Encryption Algorithm)

Blowfish

## \* Algorithm Types:

Algorithm type defines what size of plaintext should be encrypted in each step of the algorithm



# 1. Algorithm Types (2/6)

## 1) Stream Cipher:

↳ Plain text is encrypted 1 bit at a time.

Example: Plain text } Attack  
                  ASCII }

A { ASCII code  
      65

Binary representation:

Key:

Encryption Algorithm:

**A**  
01000001  
10010101  
**XOR**

**XOR:** 11010100

Cipher text: Z T U C D F (Say)

| XOR |   |
|-----|---|
| 0   | 0 |
| 0   | 1 |
| 1   | 0 |
| 1   | 1 |

# 1. Algorithm Types (3/6)

Decryption is also one bit at a time

Cipher text (binary): 1 1 0 1 0 1 0 0  
 $\oplus$  1 0 0 1 0 1 0 1  $\rightarrow$  Key

Plain text:

0 1 0 0 0 0 0 1

**XOR is reversible:** when used twice it produces the original value

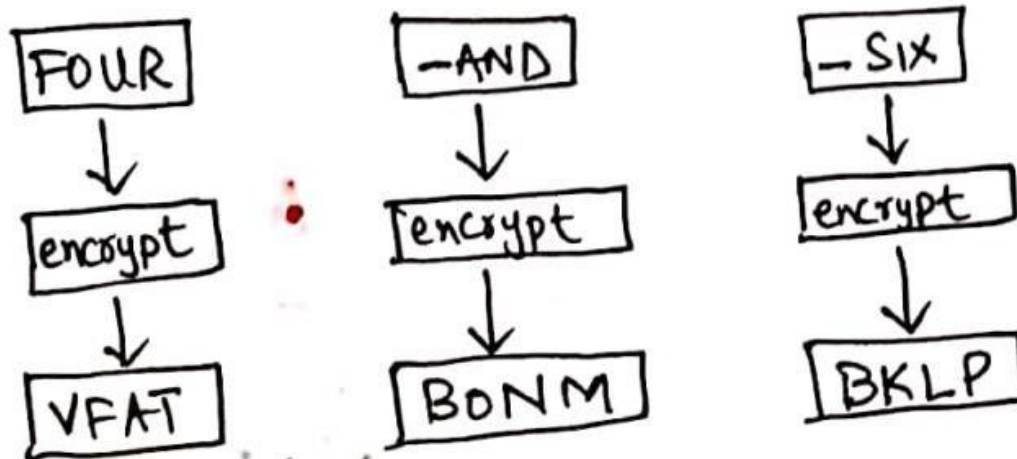
# 1. Algorithm Types (4/6)

## 2) Block Cipher:

↳ Rather than encryption one bit at a time, a block of bits is encrypted.

Example: Plaintext: FOUR -AND -SIX

Encrypt FOUR  
-AND  
-SIX



During decryption each block is translated back

# 1. Algorithm Types (5/6)

## Problem with Block Cipher

**Problem:** Repeating (plain)text patterns generate same cipher & an attacker can look for repeating strings & try to break them.

⇒ Usually the blocks used in block cipher contain 64 bits  
→ Block ciphers used often in computer-based cryptographic algorithms.



# 1. Algorithm Types (6/6)

## \* Concepts of Confusion & Diffusion:

**Confusion:** Technique of ensuring the cipher text gives no clues about original plain text.

- Try to avoid patterns in cipher text. Achieved by substitution.

**Diffusion:** Increase the redundancy of plain text by spreading it across rows & columns.

- Achieved by transposition techniques

Stream ciphers rely on confusion

Block ciphers rely on both confusion & diffusion

## 2. Algorithm Modes (1/15)

### \* Algorithm Modes:

An algorithm mode is a combination of a series of basic algorithm steps on a block cipher & some feedback from the previous step.

ECB: Electronic Code Book

CBC: Cipher Block Chaining

CFB: Cipher Feed Back

OFB: Output Feed Back

ECB }  
CBC }  $\Rightarrow$  work on block ciphers

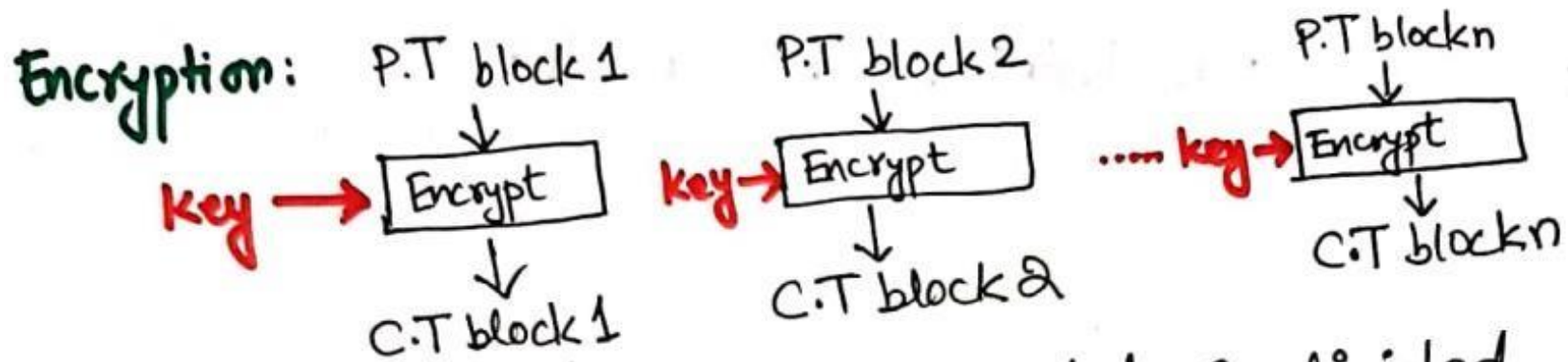
CFB }  
OFB }  $\Rightarrow$  can be used on block ciphers as if they are working on stream cipher.

## 2. Algorithm Modes (2/15)

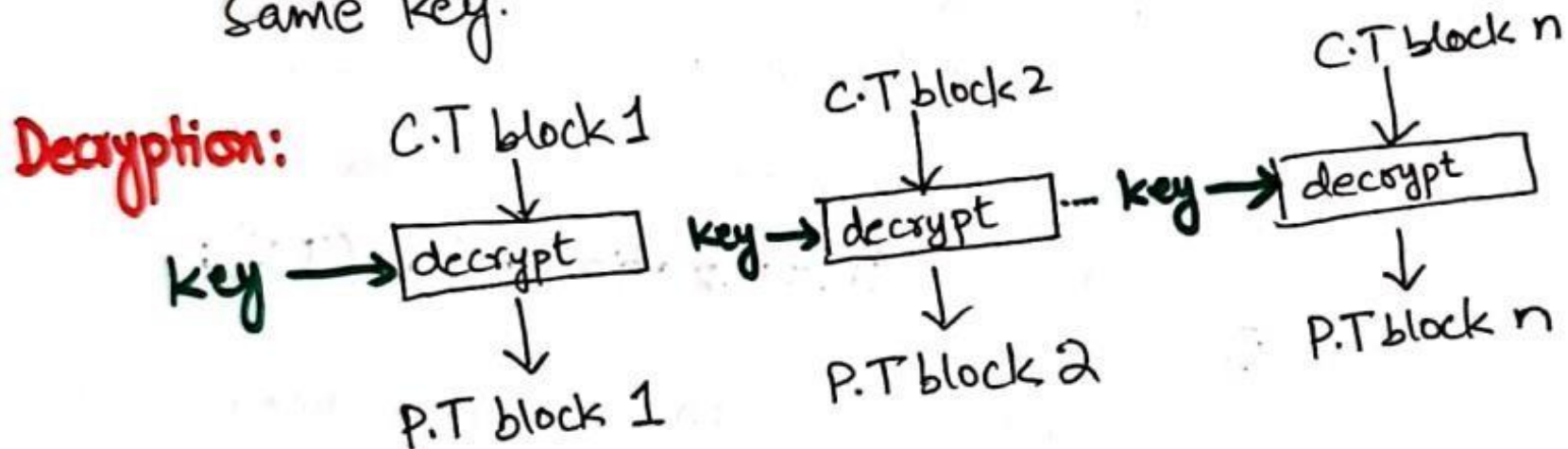
### 1) Electronic Code Book (ECB) mode:

- simplest mode
- incoming plain text message is divided into blocks of 64-bits each (Zero padding can be done if less than 64-bits)
- each block is encrypted independently
- For all blocks same key is used.

## 2. Algorithm Modes (3/15)



- At receiver end incoming data is divided into 64-bit blocks.
- each block is decrypted independently using same key.





## 2. Algorithm Modes (4/15)

### Limitation of ECB:

As a single key is used for encrypting all blocks of a message, if plaintext block repeats in the original message, the cipher text block will be also repeated in encrypted message.

⇒ ECB suitable for only encrypting small messages with repetition quite less

## 2. Algorithm Modes (5/15)

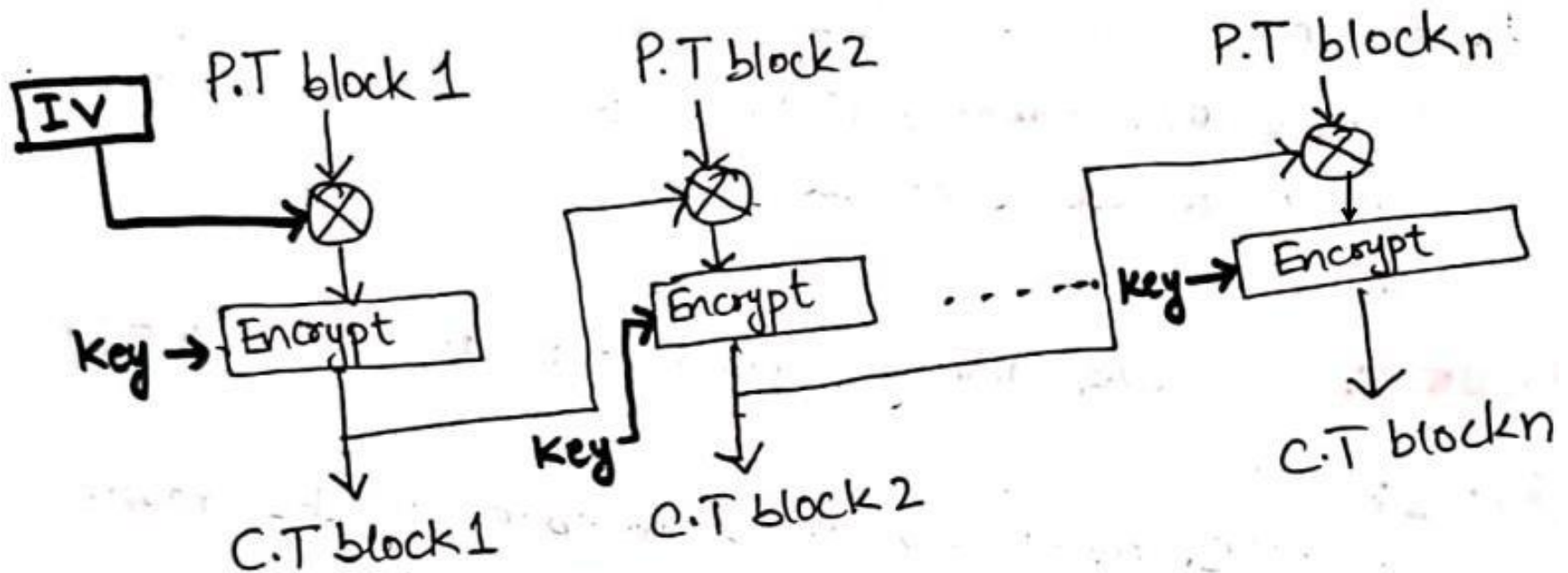
### 2) Cipher Block Chaining (CBC) mode:

- CBC mode ensures that even if a block of plaintext repeats, two (or more) identical plaintext blocks yield totally different cipher text.
- CBC uses a **feed back mechanism**.
- **chaining** adds the feed back mechanism to a block cipher.
- The results of the encryption of the previous block are fed into encryption of current block.
- Each block modifies encryption of next block.
- Initialize with a **initialization vector (IV)**: randomly generated
- Initialization vector is an arbitrary number that can be used along with secret key for data encryption.
- Both key and IV are kept secret.

## 2. Algorithm Modes (6/15)

Encryption:

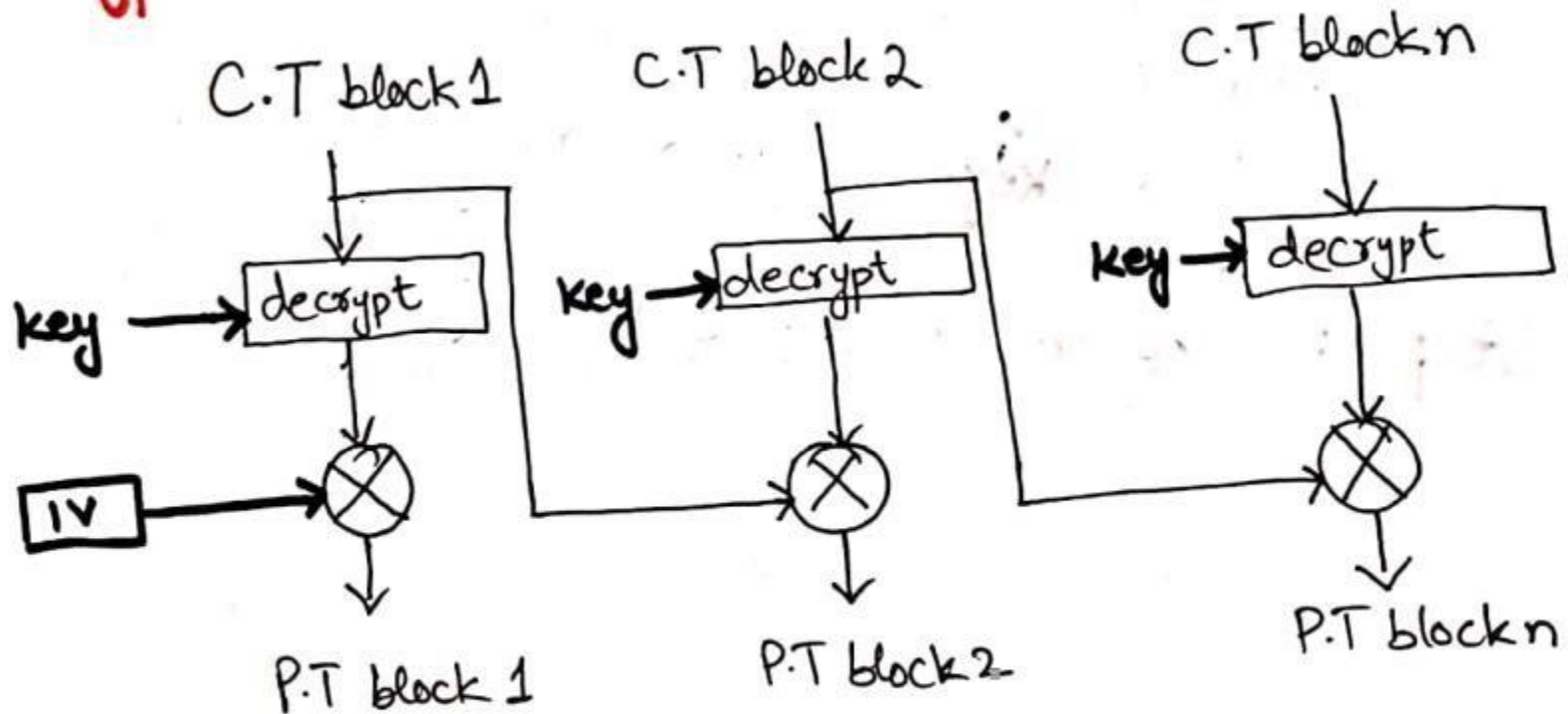
$\otimes$ : XOR



$\otimes$  = Product of two vectors/XOR Operation.

## 2. Algorithm Modes (7/15)

Decryption:



- Use same key for encryption & decryption
- Initialization Vector (IV) is used only in the first step of encryption/decryption.



## 2. Algorithm Modes (8/15)

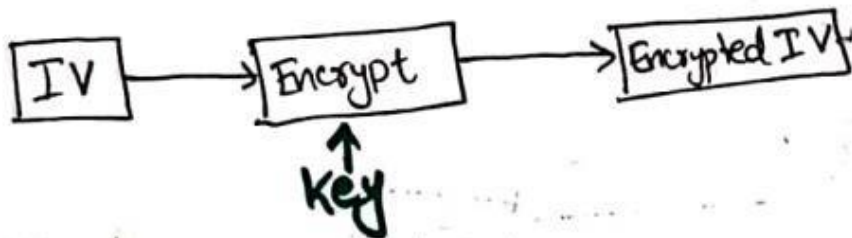
### 3) Cipher Feedback (CFB) mode:

- not all applications can work with blocks of data.
- For example: password keystrokes need to be transmitted immediately in a secure manner.
- Need to use smaller chunks e.g: 8 bits rather than a block (64 bits).

**Assume:** Dealing with ' $j$ ' bits ( $j = 8$  bits but not always)

#### Step 1:

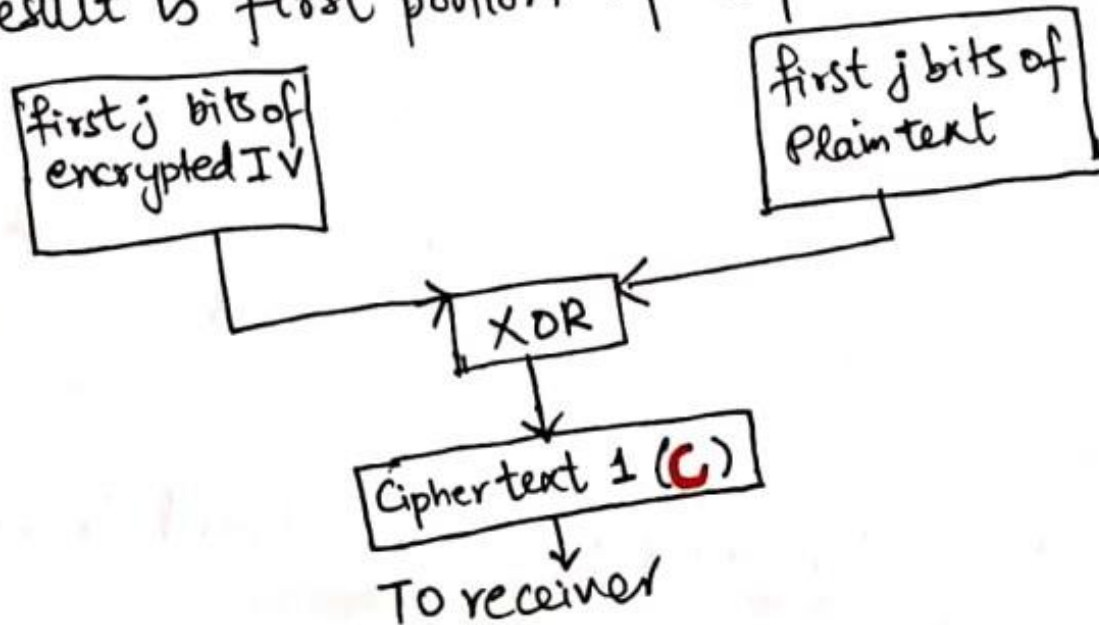
- Like CBC, an IV is used in case of CFB mode.
- IV is 64 bits & kept in a shift register.
- IV is encrypted to produce 64-bit IV ciphertext.



## 2. Algorithm Modes (9/15)

**Step 2:** Take the leftmost (most significant) 'j' bits of the encrypted IV and XOR with first 'j' bits of plain text

Result is first portion of cipher text (C):



Transmit 'C' to receiver.

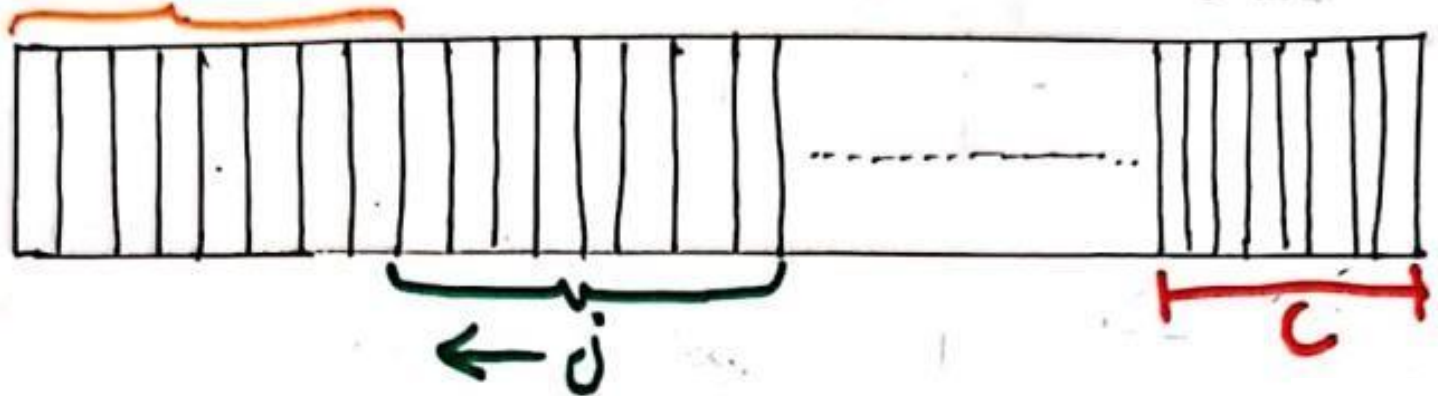
## 2. Algorithm Modes (10/15)

**Step 3:** The bits of IV (in shift register) are shifted left by ' $j$ ' positions.

The rightmost ' $j$ ' positions are filled with ' $C$ '.

$j$  (discard)

IV:



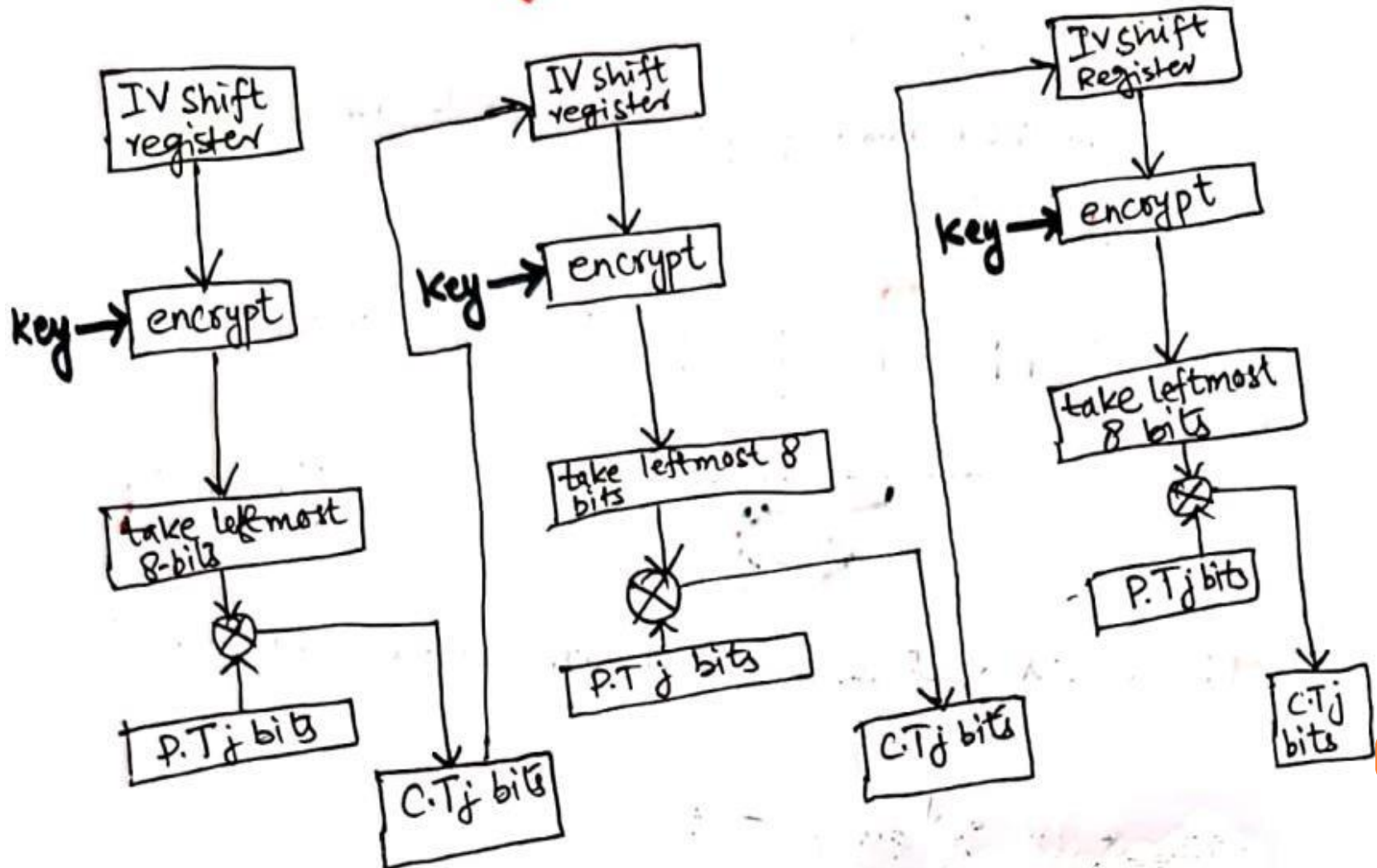
## 2. Algorithm Modes (11/15)

**Step 4:** Repeat step 1, 2, 3 till all plain text units are encrypted

- \* IV is encrypted
- \* leftmost  $j$  bits XOR with  $j$  bits of P.T
- \* resulting ciphertext ( $j$  bits) sent to receiver.
- \* Shift register containing IV is left shifted by  $j$  bits
- \*  $j$  bits of cipher text 'insert' from right into shift register

## 2. Algorithm Modes (12/15)

Overall encryption process:





## 2. Algorithm Modes (13/15)

\* Encryption

$$\rightarrow \text{Encrypted IV} \oplus P.T = C.T$$

\* Decryption

$$\rightarrow \text{Encrypted IV} \oplus C.T = P.T$$



## 2. Algorithm Modes (14/15)

### 4) Output Feed back (OFB) Mode:

→ Similar to Cipher feedback mode.

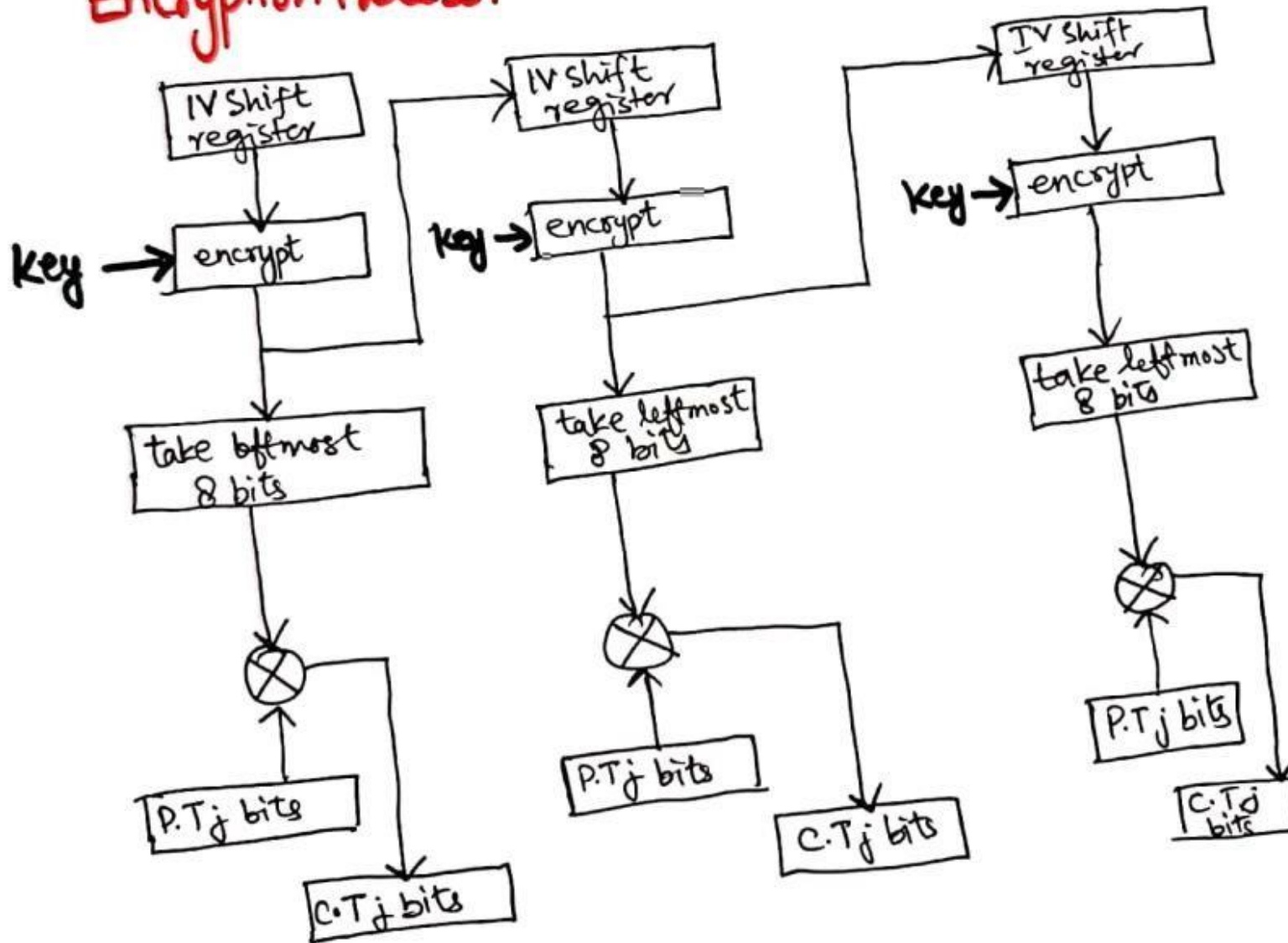
→ Only difference is:

**CFB:** Cipher text is Fed into next stage of encryption

**OFB:** Output of IV encryption process is fed into next stage of encryption process.

## 2. Algorithm Modes (15/15)

Encryption Process:



# References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2<sup>nd</sup> Edition, McGraw Hill Education (India) Private Limited.*
2. *“Cryptography and Network Security Principles and Practice”, by William Stallings, 6<sup>th</sup> Edition, Pearson.*
3. *“Computer Security Principles and Practice”, by William Stallings, Lawrie Brown, 4<sup>th</sup> Edition, Pearson.*